March 14, 2025

White House Office of Science and Technology Policy
C/O Faisal D'Souza, National Coordinating Office
Eisenhower Executive Office Building
1650 Pennsylvania Ave NW
Washington, DC 20502

Submitted electronically to: [redacted]

RE: Request for Information: Development of an Artificial Intelligence Action Plan

Dear Mr. D’Souza:

Premier Inc. appreciates the opportunity to submit comments on the Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan, issued by the National Coordinating Office for Networking and Information Technology Research and Development (NCO NITRD) on behalf of the Office of Science and Technology Policy (OSTP). Premier has deep experience and longstanding expertise with leveraging AI to reduce healthcare costs, improve quality and access, alleviate workforce shortages and advance high-quality care for communities nationwide. In our comments, Premier recommends that the Administration focus its regulatory efforts on the following:

- Publishing clear guidance on patient consent and other requirements to allow for the use of AI-generated synthetic control arms in clinical trials;
- Issuing guidance that supports the industry-wide transition to AI-powered advanced process controls and quality monitoring in drug manufacturing;
- Focusing any future federal AI algorithm transparency requirements on a standardized, outcomes-focused set of metrics instead of the “inputs” approach of “model card” method;
- Working closely with healthcare technology stakeholders to ensure that any data standards that the federal government codifies align with industry-experienced best practices;
- Requiring clear labeling of recommended use(s) to mitigate the risk that an AI algorithm will be applied in a situation for which the model is not appropriate; and
- Supporting training for the healthcare workforce that combats automation bias and incorporates human decision-making into the use of AI technology in healthcare.

Our recommendations are described in greater detail below.

I. BACKGROUND ON PREMIER INC.

Premier is a leading healthcare improvement company and national supply chain leader, uniting an alliance of 4,350 hospitals and approximately 300,000 continuum of care providers to transform healthcare. With integrated data and analytics, collaboratives, supply chain solutions, consulting and other services, Premier enables better care and outcomes at a lower cost. Premier's sophisticated technology systems contain robust data gleaned from nearly half of U.S. hospital discharges, 812 million hospital outpatient and clinic encounters and 131 million physician office visits. Premier is a data-driven organization with a 360-degree view of the supply chain, working with more than 1,400 manufacturers to source the highest quality and most cost-effective products and services. Premier's work is closely aligned with healthcare providers, who drive the product and service contracting decisions using a data driven approach to remove biases in product sourcing and contracting and assure access to the highest quality products. In addition, Premier operates the nation's largest population health collaborative, having worked with more than 200 accountable care organizations (ACOs).

Premier has a wealth of operational experience leveraging AI technology to move the needle on cost and quality in healthcare, including:

- Premier Clinical Decision Support (CDS) designs AI-enabled technology to reduce low-value and unnecessary care. Premier CDS leverages natural language processing AI technology to read unstructured data such as physician notes in electronic health records and ties it together with established practice guidelines to generate real-time alerts and relevant analytics, guiding physician's decisions at the point of care toward higher-quality, lower-cost healthcare. Premier CDS's mission is to measurably improve the quality and safety of patient care while reducing the cost of care by enabling context-specific information integrated into the provider workflow.

- Premier Applied Sciences (PAS) is a trusted leader in accelerating healthcare improvement through AI-powered solutions that span the continuum of care and enable sustainable innovation and rigorous research. Our services and real-world data drive research and quality improvement in pharmaceutical, device and diagnostic industries, academia, federal and national healthcare agencies, as well as hospitals and health systems. PAS leverages Premier's robust data resources to design and deploy AI-powered solutions for clinical trial recruitment, and to help collate disparate patient records to tell a complete patient story, leading to higher-quality care.

- Conductiv, a Premier purchased services subsidiary, harnesses AI to help hospitals and health systems streamline contract negotiations, benchmark service providers and manage spend based on historical supply chain data. Conductiv also works to enable a healthy, competitive services market by creating new opportunities for smaller suppliers and helping hospitals invest locally across many different categories of their business.

- Premier's award-winning Supply Chain Disruption Manager (SCDM) builds resilience and mitigates risks to the healthcare supply chain by harnessing machine learning AI technology to predict when critical drugs, devices and other medical supplies are anticipated to become unavailable up to six weeks in advance of a supply chain disruption. SCDM allows hospitals and health systems to access clinically approved alternative products to avoid delays in care or quality, and it allows for communication to federal agencies and other partners about pending shortages to help proactively develop mitigation strategies.

A Malcolm Baldrige National Quality Award recipient, Premier plays a critical role in the rapidly evolving healthcare industry, collaborating with healthcare providers, manufacturers, distributors, government and other entities to co-develop long-term innovations that reinvent and improve the way care is delivered to patients nationwide. Headquartered in Charlotte, North Carolina, Premier is passionate about transforming American healthcare.

II. UNLEASHING INNOVATION IN DRUG RESEARCH, MANUFACTURING AND SUPPLY CHAINS

Premier has noted a recent surge of interest in patient-facing AI technologies in clinical settings, including ambient notetaking, care navigation chatbots and AI-powered radiological consultations. However, Premier, our members and others across the healthcare sector have been using AI technology to streamline burdensome administrative processes for years. Specific use cases include:

- Harnessing the power of natural language processing (NLP) AI tools to “read” unstructured data in medical records to efficiently build medical necessity documentation to expedite prior authorization;
- Overlaying AI chatbots on enterprise resource planning software to help hospitals and health systems more efficiently identify and manage their needs; and
- Leveraging predictive AI software to sift through large, evolving datasets and proactively predict supply chain disruptions to prevent disruptions in patient care.

In addition to streamlining many administrative functions in the healthcare sector – tasks that AI has been assisting with for nearly a decade – there are many applicable use cases for AI technology in drug research, development and production.

Opportunities for AI in Clinical Trials

Premier sees particular promise for the use of AI in streamlining processes and expanding patient access in clinical trials.

Identifying trial participants: One of the biggest challenges facing health systems that seek to participate in or enroll patients in clinical trials is identifying and enrolling patients in a timely manner. Delays in meeting trial enrollment targets and timelines can increase the cost of the trial. AI tools have the ability to analyze the extensive universe of data available to healthcare systems in order to identify patients that may be a match for clinical trials that are currently recruiting. This application of natural language processing systems can make developing new drugs less expensive and more efficient, while also improving patient and geographical diversity in trials to address generalizability.

Generating synthetic data: AI, once trained on real-world data (RWD), has the capability to generate synthetic data and patient profiles that share characteristics with the target patient population for a clinical trial. This synthetic data can be used to simulate clinical trials to optimize trial designs, model the possible effects or range of results of a novel intervention, and predict the statistical significance and magnitude of effects or biases. Ultimately, synthetic patient data can help optimize trial design, improve safety and reduce cost for decentralized clinical trials. Further, synthetic control arms in clinical trials can help increase trial enrollment by easing patient fears that they will receive a placebo. ***To encourage continued innovation, Premier encourages OSTP to work with the Food and Drug Administration (FDA) on promulgating clear guidance on the process for properly obtaining consent from patients for the use of their RWD to produce AI-generated synthetic control arms in clinical trials.***

Opportunities for AI in Drug Manufacturing

Premier sees potential for AI to transform at least three key segments of the drug manufacturing process: component supply chain, advanced process control and quality monitoring.

Supply chain visibility: Premier is confident that the application of AI can advance national security by helping build a more efficient and resilient healthcare supply chain. Specifically, AI can enable better demand forecasting for products and services, such as drug components, through analysis of historical and emerging clinical and patient data. The ability to understand and react to shortages poses a critical challenge to healthcare providers; AI enables better planning and response time to national or regional emergencies. AI can drive better inventory management by automating the monitoring and replenishment of inventory levels. Healthcare providers can leverage AI to better manage suppliers through faster more efficient contracting processes and by monitoring of supplier key performance metrics. As Premier works to combat drug shortages, the most effective remedies begin with supply chain visibility and reliable predictions that allow manufacturers to plan for and respond to shortages or disruptions – this crucial element of the drug manufacturing process presents a key value-add opportunity for AI technology.

Advanced process control: Another significant role for AI in drug manufacturing is in the development and optimization of advanced process control systems (APCs). Process controls typically regulate conditions during the manufacturing process, such as temperature, pressure, feedback and speed. However, a recent report found that industrial process controls are overwhelmingly still manually regulated, and less than 10 percent of automated APCs are active, optimized and achieving the desired objective. These technologies are now ready to transform drug manufacturing on a commercial scale; however, challenges still remain to widespread adoption. ***Premier strongly believes that the FDA should issue clear guidance that supports the industry-wide transition to AI-powered APCs.*** Such technologies offer drug manufacturers the opportunity to assess the entire set of input variables and the effect of each on system performance and product quality, automating plant-wide optimization. This application of AI technology can transform the physical manufacturing of drugs and pharmaceuticals, leading to cost-savings and increased resiliency, transparency and safety in the drug supply chain.

Quality monitoring: AI can also provide value-add to drug manufacturing in the field of quality monitoring and reporting. Current manufacturing processes provide an immense volume of data from imagers and sensors that, if processed and analyzed more quickly and efficiently, could transform approaches to safety and quality control. AI models trained on this data can be used to predict malfunctions or adverse events. AI can also perform advanced quality control and inspection tasks, using data feeds to quickly identify and correct product defects or catch quality issues with products on the manufacturing line. Taken together, these capabilities can improve both the accuracy and speed of inspections and quality control, helping companies to reliably meet regulatory requirements and avoid costly delays that disrupt the drug supply chain.

III. EFFICACY, ACCURACY AND TRANSPARENCY

Premier supports the responsible development and implementation of AI tools across all segments of American industry – particularly in the healthcare industry, where numerous applications of this technology are already improving patient outcomes and provider efficiency.

Premier strongly supports AI policy guardrails that include standards around transparency and trust, risk and safety, and data use and privacy.

Promoting Transparency

Trust – among patients, providers, payers and suppliers – is critical to the development and deployment of AI tools in healthcare settings. To earn trust, AI tools must have an established standard of transparency. Some policy proposals, including those proffered by the Office of the National Coordinator for Health Information Technology (ONC), suggest transparency can be achieved through a “nutrition label” model, which lists the sources and classes of data used to train the algorithm. ***Unfortunately, some versions of the "nutrition label" approach to AI transparency fail to acknowledge that when an AI tool is trained on a large, complex dataset, and is by design intended to evolve and learn, the initial static inputs captured by a label do not provide accurate insights into an ever-changing AI tool.*** Further, overly-intrusive disclosure requirements around data inputs or algorithmic processes could force AI developers to publicly disclose intellectual property or proprietary technology, which would stifle innovation.

Premier recommends that AI technology in healthcare should be held to a standardized, outcomes-focused set of metrics, such as accuracy, false positives, inference risks and recommended use/applications. Outcomes, rather than inputs, are where AI technologies hold potential to drive health or harm. Thus, Premier believes it is essential to focus transparency efforts on the accuracy, reliability and overall appropriateness of AI technology outputs in healthcare to ensure that the evolving tool does not produce harm.

Mitigating Risks

It is important to acknowledge potential concerns around "hallucinations" and biased outcomes resulting from the use of AI tools in healthcare, which carry considerations for patient safety. Fortunately, there are several best practices that Premier and others at the forefront of technology are already following to mitigate these risks.

First, we reiterate Premier's recommendation for standardized, outcomes-based assessments of AI technologies' performance, which would hold developers accountable for reporting improper outputs. Premier also supports the development of a standardized risk assessment, which should identify detailed explanations of recommended uses for the tool and risks that could arise should the tool be applied inappropriately.

Additionally, Premier understands the importance of data standards, responsible data use and data privacy in the development and deployment of AI technology. ***Premier encourages the Administration to work closely with developers, vendors and other stakeholders to ensure that any data standards that the federal government codifies align with industry-experienced best practices.*** Premier also supports the establishment of guidelines for proper data collection, storage and use that protect patient rights and safety. This is particularly important given the sensitivity of health data.

IV. TRAINING THE HEALTHCARE WORKFORCE OF THE FUTURE

Premier believes technology can and should work alongside and learn from healthcare professionals, but current technology will not and should not replace the healthcare workforce.

To ensure clinical validity and protect patients, ***Premier recommends clear labeling of recommended use(s) and federal support for healthcare workforce trainings that combat automation bias and incorporate human decision-making into the use of AI technology in healthcare.*** Automation bias refers to human overreliance on suggestions made by automated technology, such as an AI device. This tendency is often amplified in high-pressure settings that require a rapid decision. The issue of automation bias in a healthcare setting is discussed at length by the FDA in guidance on determining if a clinical decision support tool should be considered a medical device. Premier suggests that future guidance or standards for the use of AI should consider automation bias in risk assessments and implementation practices, such as workforce education and institutional controls, to minimize the potential harm that automation bias could have on patients and vulnerable populations.

Premier acknowledges the risks of automation bias and fully automated decision-making processes. To reduce these risks, promote trust in AI technologies used in healthcare and achieve the goal of supporting the healthcare workforce through AI, ***Premier recommends that healthcare workforce training programs provide comprehensive AI literacy training***. Healthcare workers deal with high volumes of incredibly nuanced data, research and instructions – a growing percentage of which may be supplied by AI. This is particularly true for applications of AI in drug development, where manufacturers and quality control specialists may be reviewing high volumes of AI-powered recommendations or insights and making rapid decisions that affect the safety of patients. By ensuring our healthcare workers understand how to evaluate the most appropriate AI use cases and appropriate procedures for evaluating the accuracy or validity of AI recommendations, we can maximize the advisory benefit of AI while mitigating the risk to patients and provider liability.

V. CONCLUSION

Premier appreciates the opportunity to respond to this RFI on topics for the development of an OSTP AI Action Plan. If you have any questions regarding our comments, or if Premier can serve as a resource on these issues to the Administration in its policy development, please contact Mason Ingram, Senior Director of Government Affairs, at [redacted]

Sincerely,

Soumi Saha, PharmD, JD
Senior Vice President of Government Affairs
Premier Inc.